lamport robotics

Pincer1.

A new kind of cold wallet. It's a robot.

Your crypto can't move without you. Six legs, a certified secure element, and a separate controller that reads every transaction on its own display before it asks for your touch.

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What Pincer1 is

Pincer1 is a hardware wallet with a body. Every wallet today proves one thing: that someone holds the key. A seed phrase phished from a laptop, a signature clicked in a hurry, a drainer hidden in an airdrop, a phone taken from a pocket: none of these require the owner to be present. Hardware wallets fixed key storage, not the human in front of it. Trading bots fixed attention by taking the key into the cloud.

Pincer1 puts the key in a machine that can see, hear and feel. It signs only when it has verified, with its own sensors, that its owner is standing in front of it and wants this exact transaction. It reads every transaction out loud and on a display only its consent controller can write to, refuses the ones that would have robbed you, watches the market while you sleep, and tells you what happened in the morning.

One seed, every major chain: Solana, Ethereum and the EVM networks, Bitcoin. Built for traders, founders and families who hold meaningful crypto at home.

CORE FEATURES · ALWAYS ON

01

Presence-gated signing

Lidar confirms a real person in the room, the camera on the arm matches the owner's face with liveness checks from a moving viewpoint, the microphone matches a spoken passphrase against the owner's voice, and the owner touches the robot. The touch is read by a separate microcontroller wired directly to the secure element.

02

Secure element custody

Keys are generated and kept inside an NXP SE050 (CC EAL6+): ed25519 for Solana, secp256k1 for EVM and Bitcoin, from one 24-word seed on standard paths. The chip talks only to signed firmware over an encrypted channel. The PIN is a rhythm tapped on the pad and verified inside the chip; five failures wipe the device.

03

Trusted display

The consent controller has its own small display that nothing else can write to. It shows the controller's own reading of the transaction, and what you touch to approve is what that display shows. The recovery phrase appears only there, once, and is never sent over the cable to the main computer.

04

Transaction reading

The controller parses every transaction on the device before asking for consent; the robot explains it aloud: what leaves, where it goes, what is known about the destination. Hidden instructions, authority changes and unlimited approvals are called out. Daily limits are enforced on the controller and reset on its own clock.

05

Scam filter

Drainer patterns, address poisoning, fake airdrops, malicious delegations, brand-new programs and contracts. "That's a drainer. I'm not signing." Enforced on the consent controller, not on the computer that received the request.

06

Approval device

Pincer1 appears as a wallet next to Phantom on Solana apps, next to MetaMask on EVM apps, and as a Bitcoin wallet on PSBT apps, through a browser extension. The extension is untrusted input; the controller re-reads everything it forwards.

07

Duress and decoy

A hidden word, or a second tap rhythm used instead of your PIN, opens a decoy wallet with its own 24 words, locks the real one and alerts silently. The duress rhythm is checked inside the chip and never counts as a failed attempt. You fund the decoy at setup with a believable balance.

08

Companion and brain

It talks, listens, remembers and walks over when called. The brain is your choice: Lamport-8B, our own crypto-specific model running on the robot, or Claude, ChatGPT or DeepSeek through their APIs. Whichever you pick, it only talks; no security decision comes from it.

Security model

COMPONENT	TRUSTED?	WHY
SE050 secure element	yes	keys generated and used inside a certified chip; cannot be read out
Consent controller (RP2350)	yes	only thing that can ask the chip to sign; secure boot, signed firmware; re-parses every transaction; touch on a pin only it can read; owns the trusted display
Main computer (Jetson)	no	network, RPC, models, agent, the front screen; assumed compromisable
Browser extension	no	untrusted input
The owner	final authority	the robot warns; it does not overrule a correctly warned owner

What follows: no touch, no signature. The controller refuses when the main computer's claims understate the bytes, and refuses a Solana destination it cannot reconcile with the bytes it parsed. A swapped controller cannot talk to the chip. Physical seizure is covered by secure boot, in-chip PIN and the certified part. Not covered: a stolen recovery phrase, and an owner who approves a correctly warned transaction. Everything is open source so the claims can be read rather than trusted.

Optional features off until the owner switches them on

FEATURE	WHAT IT DOES
Room lock	Signs only in its enrolled room (lidar fingerprint or map position); elsewhere it refuses.
Two-person rule	Above an amount you choose, two enrolled people must be in the room.
Offline payments	Signs durable-nonce transactions with no network; QR or deferred broadcast.
Air-gap relay	The radio is physically switched by a relay the robot controls, by presence or an agent window you authorised.
Dead-man's switch	Heir and a number of days; the timer and the signed transaction live on the controller; warns in person and by phone; nothing on the main computer can trigger it early or cancel it.
Wake-up alarm	"If my holdings drop 15%, wake me." Walks to your room and gets louder until you touch it; phones you if you are away. Quiet hours are yours to set.
Morning briefing	What moved overnight in the assets you follow, the news you asked for, what the agent did and what it proposes. Spoken when you come down, with every number read exactly as fetched.
Overnight agent	A budget granted by a long touch on the controller, bounded execution, automatic expiry, and a morning report for anything larger.
Modes and swaps	Offline mode: radio off, Lamport-8B only. Online mode: any brain, the agent, and swaps only through routers on the controller's allowlist, with the minimum output verified before you touch.
Mapping and navigation	Maps the home with lidar; you name rooms; it goes to any of them.
Charging dock	Returns to charge below 20%, leaves at 90% or when needed.
Locator beacon	Own battery, GPS, Find My over BLE and LoRa to your base station; reports where it is if carried off.

Hardware

SUBSYSTEM	PART
Body	Hexapod chassis, serial-bus servos with position feedback, aluminium plates
Arm and head	4-axis arm carrying a stereo depth camera (Luxonis OAK-D Lite)
Front display	4.3-inch IPS panel driven by the main computer: conversation, the spoken reading, status. Never the recovery phrase.
Consent module	Raspberry Pi Pico 2 (RP2350) + NXP SE050 + capacitive touch pad + its own OLED display + radio relay
Main computer	NVIDIA Jetson Orin Nano Super 8 GB, encrypted NVMe, secure boot; all models run on the robot
Sensors	360° 2D lidar, IMU, 4-microphone array, speakers
Power	3S Li-ion 21700 pack, about 200 Wh: 4–5 h walking and talking, 10 h+ standing watch

How it compares

	PHONE WALLET	HARDWARE WALLET	PINCER1
Where the key lives	phone memory	secure element	secure element
What approves a transaction	a tap	a button	a verified person in the room plus a touch
Knows if the owner is present	no	no	yes: lidar, face, voice
Reads the transaction to you	partially	tiny screen	aloud, with the counterparty's history, on a display only the controller writes
Refuses drainers on its own	no	no	yes
Under duress	nothing	some: a hidden PIN	word or rhythm opens a funded decoy and alerts silently
Works offline	no	signs only	signs and keeps a valid transaction for later
Works while you sleep	no	no	yes, inside a touch-granted budget
Inheritance	no	no	built in, enforced on the controller
Wakes you when the market moves	no	no	yes

Status and roadmap

The whole product is one open-source repository: firmware, daemon, robot software, browser extension, locator beacon and the training pipeline for Lamport-8B. Every component is written and tested in software against simulated hardware. Since the first brief, the consent path has gained its own display, the duress rhythm and the decoy, daily limits that reset on the controller's clock, allowlisted swap routers, offline and online modes, Bitcoin and EVM signing, and the morning briefing.

What remains is the hardware on the bench. Next, in order: the consent module running on real parts, the first "no signature unless you're in the room" demo on video, the approval device in a live browser, the overnight agent and morning briefing on the finished body, the optional features tuned on hardware, an independent review of the consent path, then the first production run.

LAMPORT ROBOTICS

A lamport is the smallest unit of SOL. The name also nods to Leslie Lamport's work on distributed systems. We build physical machines around digital ownership.

SITE
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NXP's SE050 component certification is not certification of the complete Pincer1 product. No filter guarantees detection of every harmful transaction. Ordering is not open yet; details will be announced at lamportrobotics.com/presale.